

Software Requirements Specification (SRS)

OpenCart E-Commerce Website

Industry Style QA Practice Document

1. Introduction

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for an OpenCart-based e-commerce application. It is intended as a realistic practice document for Manual QA Engineers.

2. Purpose

Provide a detailed baseline for requirement analysis, test planning, test case design, regression testing, and defect reporting.

3. Scope

The application allows customers to browse products, register accounts, manage shopping carts, place orders, manage profiles, and interact with customer support.

4. Stakeholders

Customer, Administrator, Warehouse Staff, QA Engineer, Business Analyst, Developer, Product Owner.

5. User Roles

Guest: Browse products, search, register.

Registered Customer: Purchase products and manage account.

Administrator: Manage catalog, customers, orders, reports and settings.

6. Business Rules

Users must register before using Wishlist. Products cannot be ordered beyond available stock unless backorders are enabled. Order confirmation is generated only after successful payment.

7. Functional Requirements

Homepage

FR-001: Display logo, menu, featured/latest products, promotional banners, footer links.

FR-002: Support navigation to all categories.

FR-003: Display search bar.

Registration

FR-004: Mandatory First Name, Last Name, Email, Telephone, Password and Privacy Policy acceptance.

FR-005: Email must be unique.

FR-006: Password confirmation must match.

Login

FR-007: Authenticate valid users.

FR-008: Display appropriate errors for invalid credentials.

FR-009: Support Forgot Password.

Search

FR-010: Keyword search.

FR-011: Category filter.

FR-012: Description filter.

Category

FR-013: List products with sorting and pagination.

FR-014: Display breadcrumbs.

Product Details

FR-015: Images, gallery, price, stock, description, specifications, reviews.

Wishlist

FR-016: Add/Remove products.

FR-017: Require authentication.

Compare

FR-018: Compare multiple products.

Shopping Cart

FR-019: Add, update quantity, remove items.

FR-020: Calculate subtotal, tax, shipping and total.

Checkout

FR-021: Billing details.

FR-022: Shipping method.

FR-023: Payment method.

FR-024: Order confirmation.

Orders

FR-025: View order history.

FR-026: View order details.

My Account

FR-027: Edit profile.

FR-028: Manage addresses.

FR-029: Newsletter.

FR-030: Password change.

Reviews

FR-031: Submit ratings and comments.

FR-032: Validate mandatory fields.

Contact Us

FR-033: Submit enquiry.

FR-034: Validation on required fields.

Admin

FR-035: Manage products.

FR-036: Manage categories.

FR-037: Manage customers.

FR-038: Manage orders.

FR-039: Reports.

8. Non-Functional Requirements

NFR-01: Application shall support Chrome, Edge and Firefox.

NFR-02: Major pages should load within acceptable response time under normal load.

NFR-03: All sensitive pages shall use HTTPS.

NFR-04: Passwords shall be securely stored.

NFR-05: System shall provide meaningful validation messages.

NFR-06: Application shall be available 24x7 except scheduled maintenance.

NFR-07: UI shall be responsive for desktop and tablet.

9. Acceptance Criteria

Homepage: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Registration: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Login: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Search: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Category: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Product Details: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Wishlist: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Compare: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Shopping Cart: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Checkout: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Orders: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

My Account: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Reviews: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Contact Us: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

Admin: All mandatory workflows execute successfully, validation behaves correctly, navigation is consistent, and no critical defects remain open before release.

10. Suggested QA Deliverables

- Requirement Traceability Matrix (RTM)
- Master Test Plan
- Module-wise Test Scenarios
- 300-500 Test Cases
- Smoke Test Suite
- Regression Test Suite
- Bug Report
- Daily Status Report
- Test Execution Report
- Test Summary Report

11. Recommended Test Modules

- Homepage
- Registration
- Login
- Search
- Category
- Product Details
- Wishlist
- Compare
- Shopping Cart
- Checkout
- Orders
- My Account
- Reviews
- Contact Us
- Admin

Appendix A: Practice Guidance

Homepage

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Registration

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Login

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Search

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Category

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Product Details

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Wishlist

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Compare

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Shopping Cart

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Checkout

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Orders

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

My Account

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Reviews

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Contact Us

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.

Admin

Create positive, negative, boundary value, validation, usability, compatibility and security-oriented test cases for this module. Aim for 20-40 test cases per module.